Solutions:

Q1: (a) $v_o=(1)(8)=8$ V.

(b) $i_b=(50)(0.2)=10$ A. $\cdot p_{8\Omega}=(8)2^8=(1)2(8)=8$ W

(c) $v_{20\Omega}=(1)(20)=-20$ V. $\cdot p_{50\Omega}=(50)^2(0.2)=500$ W

(d) $i_d=-50/25=-2$ A. $\cdot p_{20\Omega}=(1)^2(20)=20$ W

$\cdot p_{25\Omega}=(-2)^2(25)=100$ W

Q2: $-10+6i_o=0$ and $-3i_o+2i_o+3i_o=0$

Solving for the currents yields

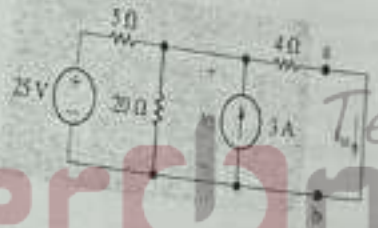
$i_o=1.67$ A and $i_o=1$ A.

Applying Ohm's law to the 3 Ω resistor gives the desired voltage:

$v_o=3i_o=3$ V.

Q3: Using nodal voltage method:

$(v_1-25)/5+v_1/20-3=0.$ ($v_1=32$ V$=V_{Th}$)



$(v_2-25)/5+v_2/20-3+v_2/4=0.$

$v_2=16$ V.

Hence, the short-circuit current is

$I_{sc}=16/4=4$ A.

$R_{Th}=V_{Th}/I_{sc}=32/4=8\ \Omega$

Q4:

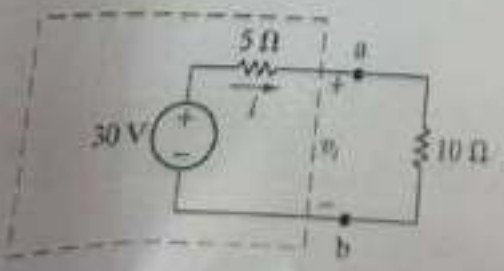
From A to C: R_3 is in parallel with the series combination of R_1 and R_2.

$$R_3 \parallel (R_1 + R_2)$$

From B to C: R_2 is in parallel with the series combination of R_1 and R_3.

$$R_2 \parallel (R_1 + R_3)$$

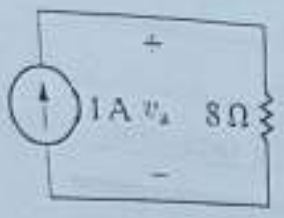
Q5:



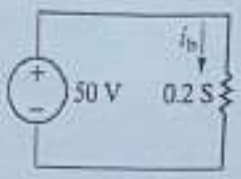
a)

b) $p_{10\Omega}=(2)2(10)=40$ W.

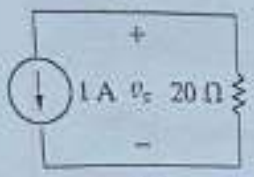
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(a)



(b)



(c)



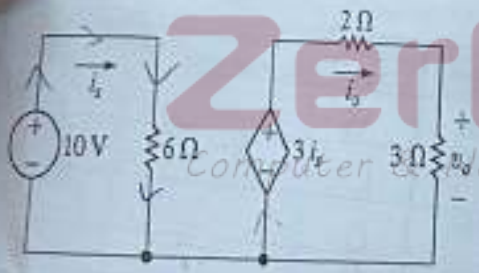
(d)

Q1: Calculate the following values for the above circuits: **(8 marks)**

(a) v_a=... ,$p_{8\Omega}$=...
(b) i_b=... ,$p_{0.2S}$=...
(c) v_c=... ,$p_{20\Omega}$=...
(d) i_d=... ,$p_{25\Omega}$=...

+5

Use KVL& Ohm's law to find the voltage v_o **(4 marks)**



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Solution:

$V = IR$

$10 = i_s \times 6$ (+1)

$i_s = \frac{10}{6}$ A

$3i_s = 5$ by KCL

$3i_1 = i_o = 5$ A

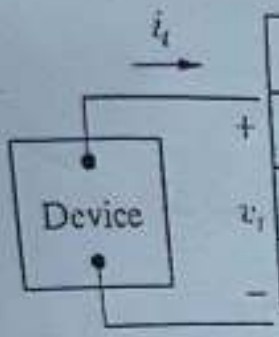
KVL $-5 - 5(2) + 3v_o = 0$

$v_o = i_o R$

$= -5 \times 3 = -15$ V

Q5:

(4 marks)

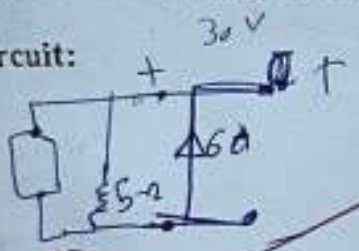


v_t (V)	i_t (A)
30	0
15	3
0	6

(a) (b)

1. Construct a circuit model of the device inside the box.
2. Using this circuit model, predict the power this device will deliver to a10 Ω resistor.

1. **The circuit:**



2. $p = i^2 R = 36 \times 10 = 360\ W$

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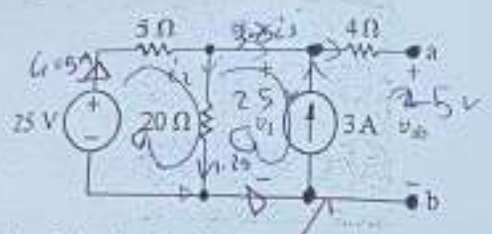
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Q3: Find the Thevenin equivalent voltage of the circuit in Fig and calculate R_{TH} using: $V_{TH}/I_{s.c}$

(10 marks)



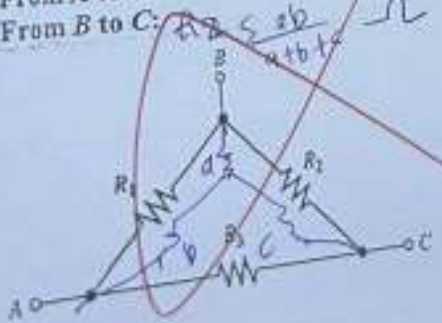
1. Calculating V_{TH}:

2. Finding R_{TH}, using $R_{TH} = V_{TH} / I_{s.c}$

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Q4: Describe the total resistance between each pair of terminals in Figure

1. From *A* to *C*:
2. From *B* to *C*:



(4 marks)